

OPENROCKET ROCKET DESIGN ASSIGNMENT

Design and Simulation of a Single-Stage Model Rocket Using OpenRocket



INDIA SPACE LAB

ProjectReport

2ndFebruaryto3rdMarch2026

StudentName: Annie Darling Kanmani

InstituteName: SRM Madurai college for Engineering and Technology

InstituteRollno.: 911123243007

Enrollmentno.: ISL-408242

OPENROCKET ROCKET DESIGN REPORT

1. Introduction

This project focuses on the design, simulation, and optimization of a single-stage model rocket using OpenRocket software. The objective of the assignment was to design a rocket that meets the specified mission requirements while ensuring aerodynamic stability and safe recovery. The rocket must reach a minimum apogee of 1000 meters, maintain a stability margin between 1.0 and 2.0 calibers, and ensure a descent rate below 10 m/s for safe landing. Additionally, the rocket design must satisfy structural and performance constraints such as a total length between 800 mm and 1200 mm, a fixed body diameter of 54 mm, and a maximum lift-off mass of 1.5 kg.

The design process involved selecting appropriate rocket components including the nose cone, body tube, fins, propulsion system, avionics payload, and recovery system. After assembling the rocket structure in OpenRocket, flight simulations were performed to verify the rocket's performance and stability characteristics.

2. Rocket Design Configuration

The designed rocket follows a single-stage configuration with a cylindrical body tube and aerodynamic nose cone. The rocket has a total length of approximately 900 mm and a fixed body diameter of 54 mm, which satisfies the design constraints provided in the assignment. The structural material selected for the rocket body is cardboard, while the fins are assumed to be made of plywood for sufficient strength and durability.

The rocket is composed of the following main subsystems:

- Nose Cone
- Body Tube
- Fin Set
- Motor Mount and Propulsion System
- Avionics and Payload
- Recovery System

Each subsystem was carefully selected and designed to ensure optimal aerodynamic performance and flight stability.

3. Nose Cone Design

The nose cone plays a crucial role in reducing aerodynamic drag and improving flight efficiency. For this design, an **Ogive nose cone** was selected. The ogive shape

provides smoother airflow over the rocket body and reduces pressure drag during flight compared to simpler shapes such as conical nose cones.

The nose cone has a length of approximately 200 mm and a diameter of 54 mm. This configuration ensures a streamlined shape while maintaining compatibility with the body tube diameter. The ogive shape is commonly used in model rockets due to its good balance between aerodynamic performance and manufacturability.

4. Fin Design

The rocket uses a **three-fin trapezoidal configuration** to provide aerodynamic stability during flight. Fins are essential for maintaining directional stability and ensuring that the rocket flies along its intended trajectory.

The fin dimensions used in this design are:

- Root Chord: 120 mm
- Tip Chord: 60 mm
- Fin Height: 80 mm
- Number of Fins: 3

The trapezoidal fin shape was selected because it provides good aerodynamic stability while maintaining structural simplicity. The fins are positioned near the rear of the rocket to shift the **Center of Pressure (CP)** behind the **Center of Gravity (CG)**. This configuration ensures that the rocket remains stable throughout its flight.

5. Propulsion System

The propulsion system selected for this rocket is a **commercial H-class rocket motor**, specifically an H128 motor. The motor was chosen to provide sufficient thrust to lift the rocket while meeting the thrust-to-weight ratio requirement of greater than 5:1 at lift-off.

The motor mount has a diameter of 29 mm and is located in the lower section of the rocket body tube. The selected motor provides adequate thrust and burn duration to achieve the target apogee of more than 1000 meters while keeping the rocket within the subsonic flight regime with a maximum Mach number below 0.8.

6. Avionics and Payload

The rocket includes a payload mass of **75 grams**, which satisfies the payload requirement specified in the assignment. In addition to the payload, an **altimeter** is included in the avionics section. The altimeter is used to detect the apogee of the rocket and trigger the recovery system deployment.

Including avionics components such as an altimeter is essential for monitoring flight performance and ensuring accurate parachute deployment during the recovery phase.

7. Recovery System

A **parachute recovery system** is used to ensure safe descent and landing of the rocket after reaching apogee. The parachute is deployed automatically when the rocket reaches its highest altitude. A shock cord is also included to connect the rocket components and absorb the forces generated during parachute deployment.

The parachute size was selected to maintain a descent rate below **10 m/s**, ensuring safe recovery without damaging the rocket structure.

8. Flight Simulation Results

The rocket design was simulated using OpenRocket to verify its performance characteristics. The simulation results confirm that the rocket meets all mission requirements.

Key simulation results include:

- Apogee: Greater than 1000 meters
- Stability Margin: Between 1.0 and 2.0 calibers
- Maximum Velocity: Within safe limits and below Mach 0.8
- Descent Rate: Less than 10 m/s
- Proper CG and CP separation ensuring stable flight

The simulation also provided detailed flight profiles including velocity, acceleration, and altitude graphs, which were used to evaluate the rocket's performance.

9. Conclusion

The designed rocket successfully meets all the mission requirements specified in the assignment. The selected ogive nose cone reduces aerodynamic drag, while the trapezoidal fin configuration ensures stable flight. The H-class motor provides sufficient thrust to achieve the required apogee, and the parachute recovery system ensures safe landing.

Through simulation in OpenRocket, the rocket demonstrated stable flight characteristics with an appropriate stability margin and acceptable descent rate. Overall, the design effectively balances aerodynamic efficiency, structural simplicity, and flight safety.